Bank Customer Churn Prediction

Menna Shafiek Mohamed
Alaa Abobakr Khalifa
Tasneem Abdelrahman
Maghfra Ashraf
Norhan Atef Mostfa

The Business Problem

Why customer churn matters to a retail bank

Losing a customer costs far more than gaining one.

- Acquiring a new bank customer typically costs 5–7x more than retaining an existing one
- Early identification of at-risk customers lets retention teams intervene before the account closes
- This project builds a predictive model that flags customers likely to churn, using their profile and account behavior
- Goal: turn raw transactional data into a ranked, actionable churn-risk score

PROJECT PIPELINE

1. Data Understanding & Cleaning
2. Exploratory & Statistical Analysis
3. Feature Engineering & Preprocessing
4. Model Training & Tuning
5. Evaluation & Business Recommendations

Dataset Overview

Bank Customer Churn Prediction dataset

10,000	12	0	0
customer rows	raw features	missing values	duplicate rows

KEY FEATURES

- credit_score
- country, gender, age
- tenure (years with bank)
- balance (account balance)
- products_number
- credit_card (owns one?)
- active_member (engaged?)
- estimated_salary

Target variable: churn — 1 if the customer closed their account, 0 otherwise

Data Understanding & Cleaning

Preparing a reliable foundation for analysis

Structure check

Inspected shape, dtypes, and summary statistics with .info() and .describe()

Quality check

Confirmed zero missing values and zero duplicate rows across all 10,000 records

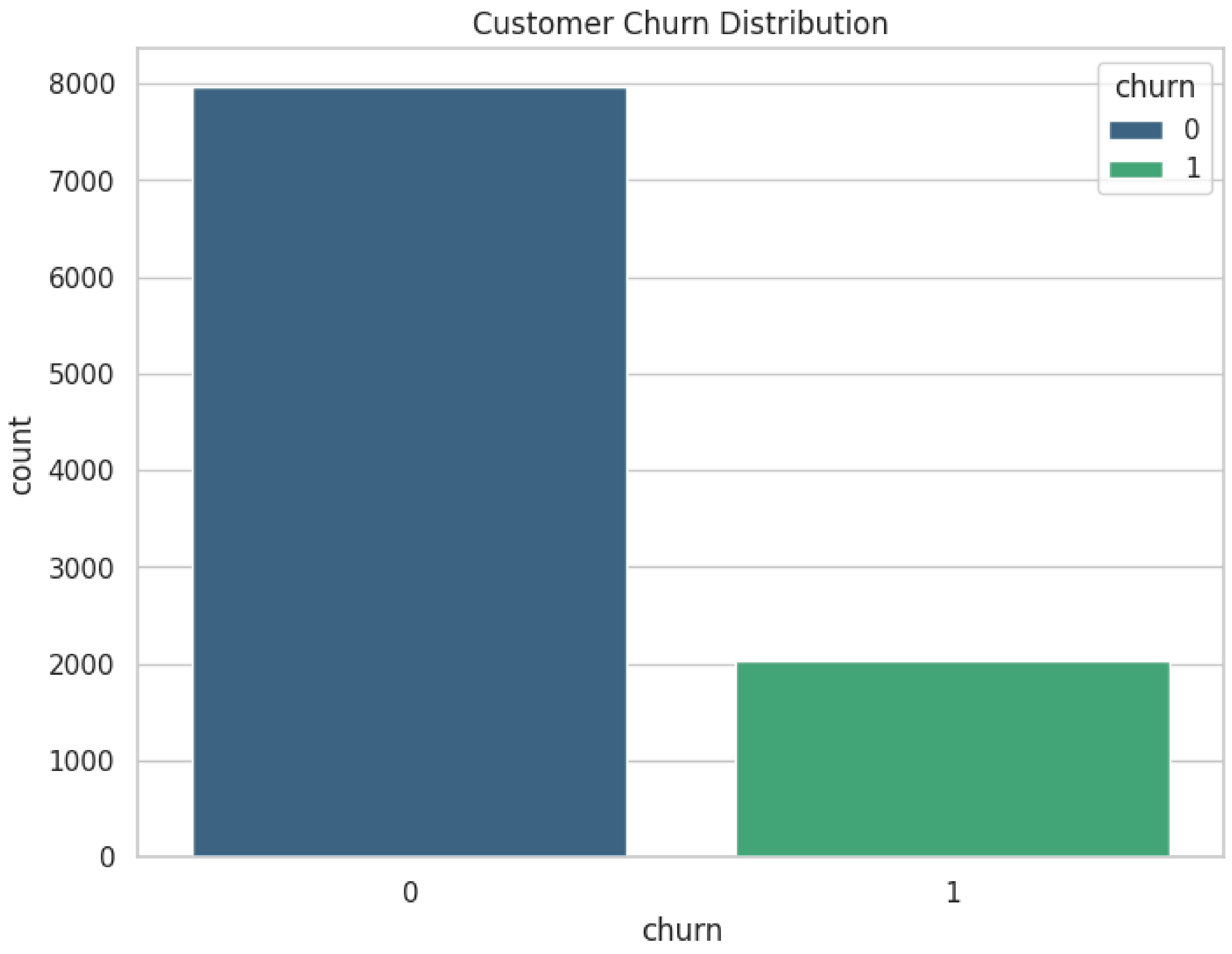
Column pruning

Dropped customer_id — a unique identifier with no predictive value

Result: a clean, complete 10,000 x 11 dataset — no imputation or row removal needed before analysis.

An Imbalanced Target

Churn distribution across the customer base

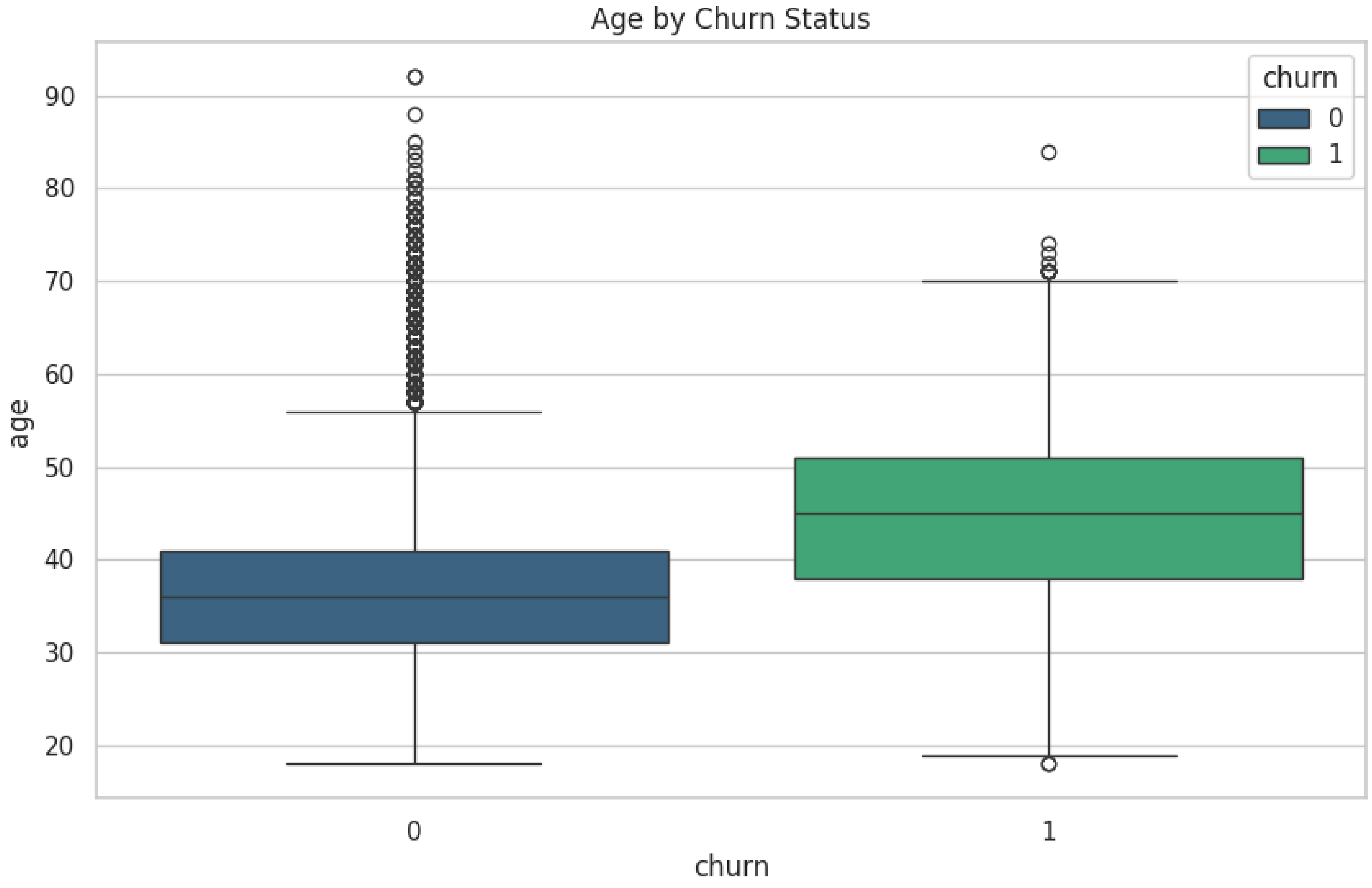


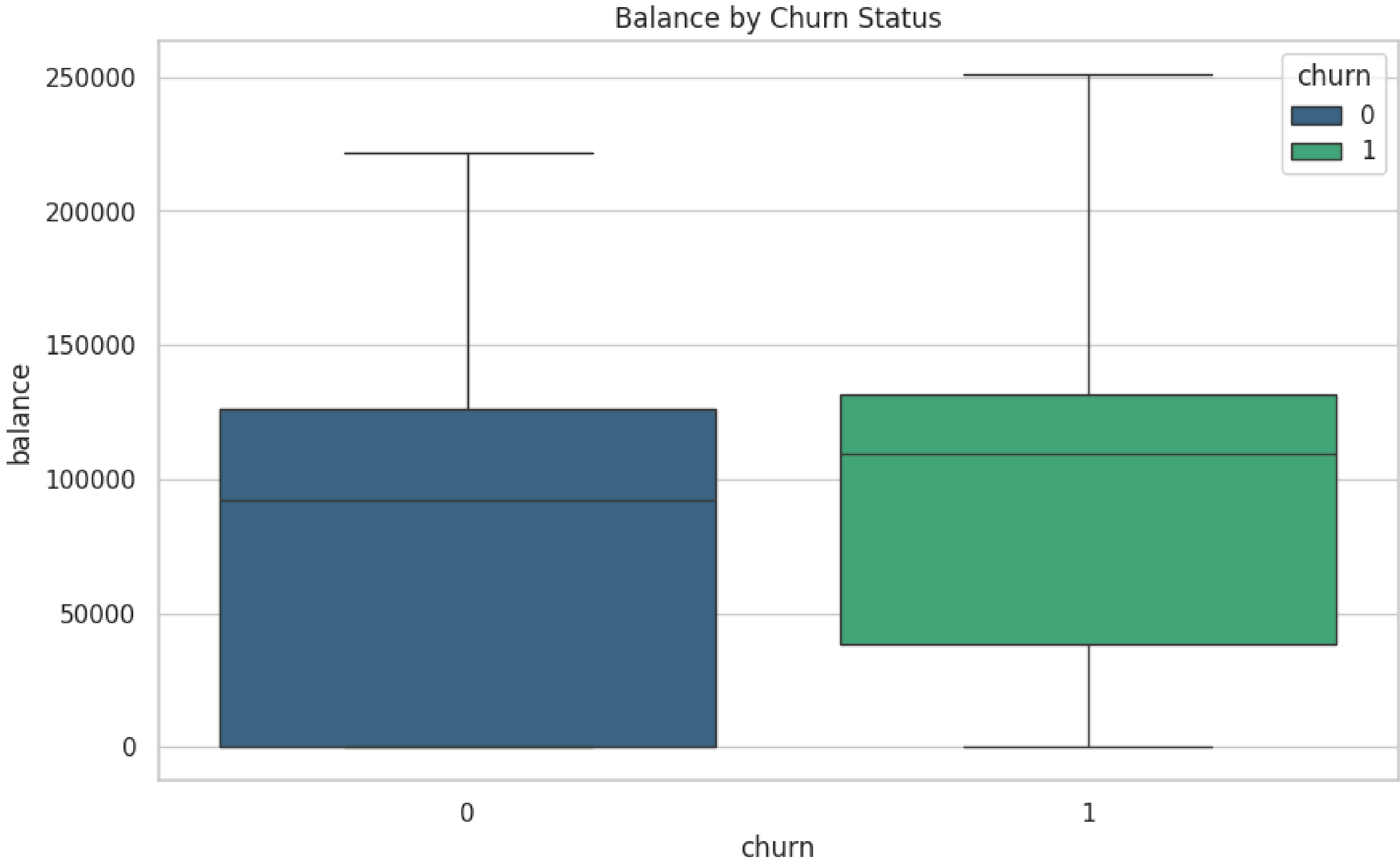
Class imbalance

- ~80% of customers stayed; only ~20% churned
- A naive model could hit 80% accuracy by predicting 'no churn' every time — accuracy alone is misleading here
- This motivates using F1-score and ROC-AUC as the real success metrics, and applying SMOTE to rebalance the training data

Who Is Churning? — Age & Balance

Bivariate analysis against churn status

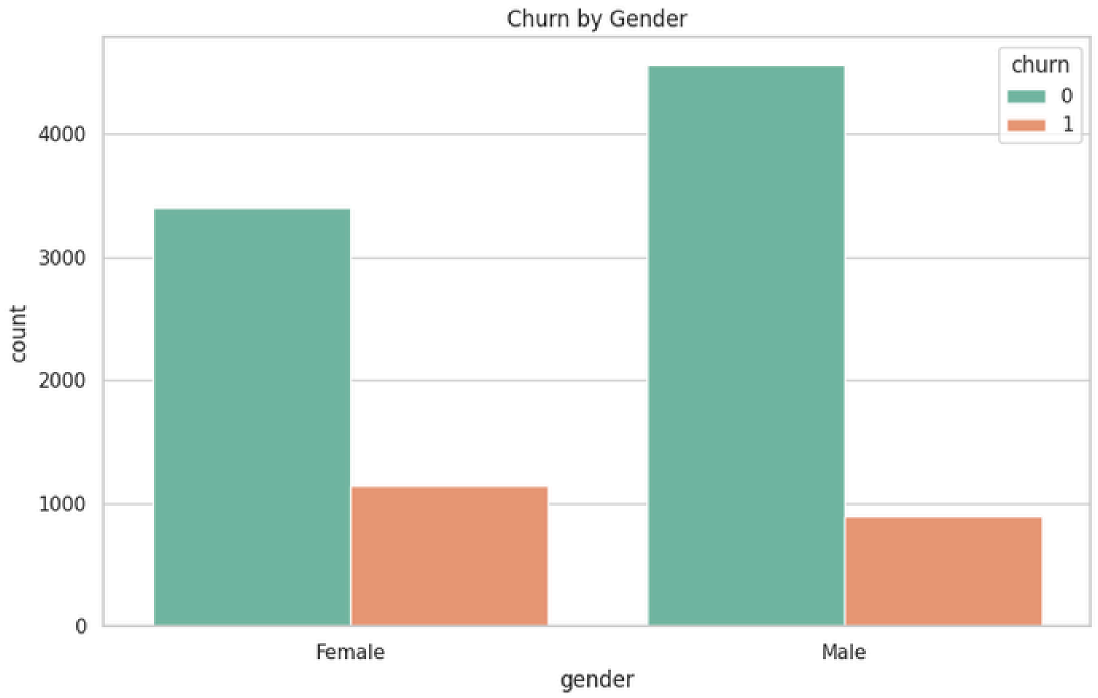




Churned customers skew older (median 45 vs 38) and carry **slightly higher balances** — age is the strongest single driver identified in the correlation analysis.

Who Is Churning? — Gender, Geography & Engagement

Categorical drivers of churn behavior



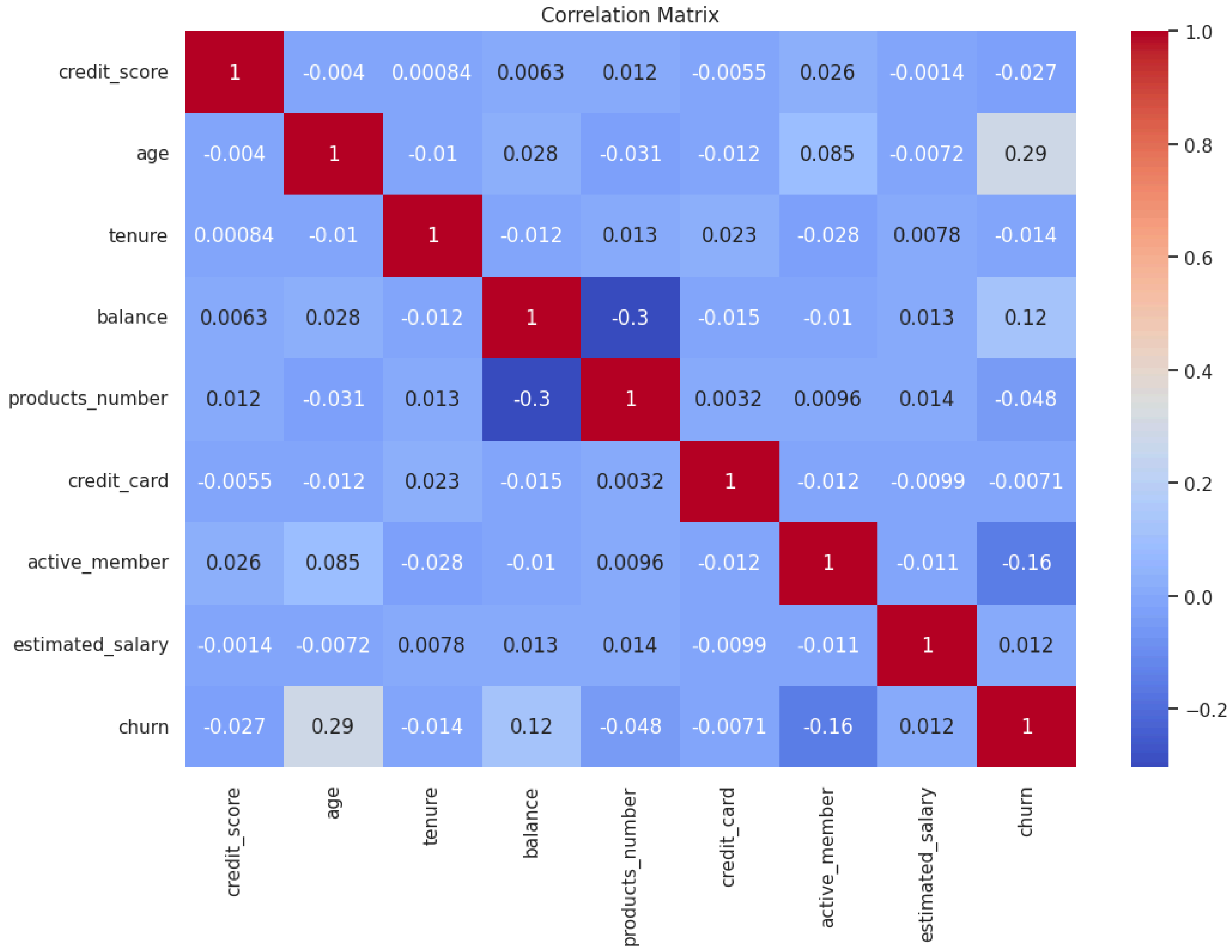
Germany

has the highest churn proportion of any country, despite France having the largest customer base overall.

- Female customers churn at a higher rate than male customers, despite men being the larger group
- Inactive members churn far more than active members — engagement is protective
- Both associations are statistically significant (chi-square $p < 0.05$)

Correlation Matrix

How numeric features relate to churn and each other



TOP DRIVERS OF CHURN

Age
+0.29

strongest positive correlation

Active member
-0.16

engaged customers churn less

Balance
+0.12

weak positive correlation

Correlations are individually weak — confirming that churn is driven by an interaction of factors, not one dominant variable.

Statistical Validation

Confirming which differences are real, not random

Test	Variable	Statistic	p-value	Result
Chi-square	Gender	112.92	2.25e-26	Significant
Chi-square	Active member	242.99	8.79e-55	Significant
T-test	Age	29.77	1.24e-186	Significant
T-test	Balance	11.94	1.28e-32	Significant
T-test	Credit score	-2.71	0.0067	Significant
T-test	Products (1 vs 3+)	2.15	0.032	Significant
ANOVA	Products group vs Balance	4.61	0.032	Significant

Every tested variable shows a statistically significant relationship with churn ($p < 0.05$) — validating the EDA findings and confirming these features carry real predictive signal, not noise.

Feature Engineering & Preprocessing

Preparing the data for machine learning

Encoding & Scaling

- Encoded gender (binary) and one-hot encoded country
- Scaled numeric features with MinMaxScaler
- Removed remaining duplicate rows after encoding

Feature Engineering

- Engineered products_group (1 / 2 / 3+ products)
- Created balance_salary_ratio and high_balance indicator
- Selected top 6 features via Recursive Feature Elimination (RFE)

RFE selected: credit_score, gender, age, balance, active_member, country_Germany

Model Training Strategy

Handling imbalance and comparing candidate algorithms

SMOTE oversampling

Balanced the training set (Synthetic Minority Over-sampling) so models learn churn patterns, not just the majority class

Three candidate models

Logistic Regression, Random Forest, and Gradient Boosting — trained on the SMOTE-balanced training data

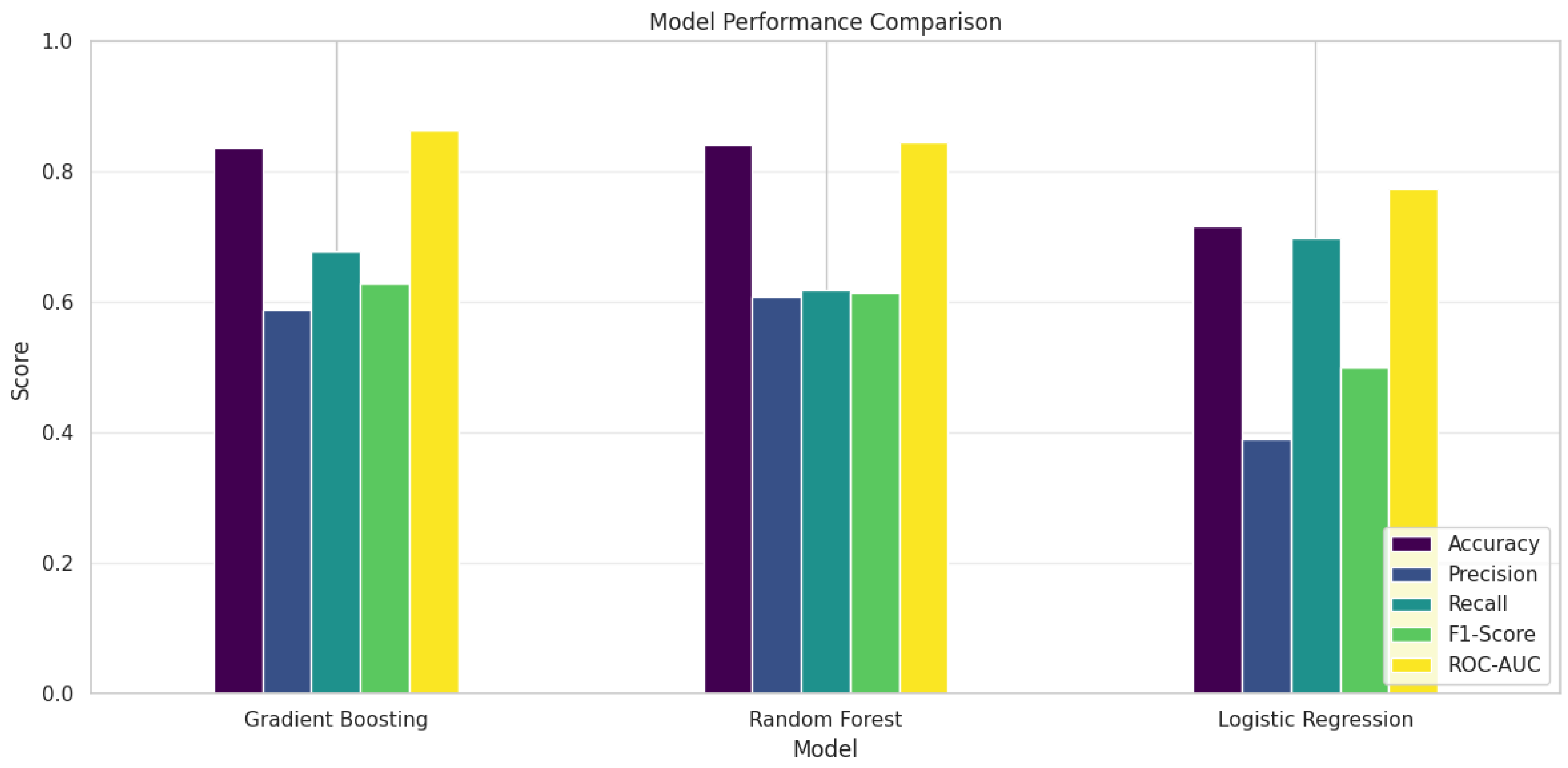
5-fold cross-validation

Stratified CV using F1-score to fairly compare models under class imbalance

Cross-val F1 (SMOTE-balanced training): Random Forest 0.898 | Gradient Boosting 0.866 | Logistic Regression 0.709

Model Performance Comparison

Evaluated on the held-out test set



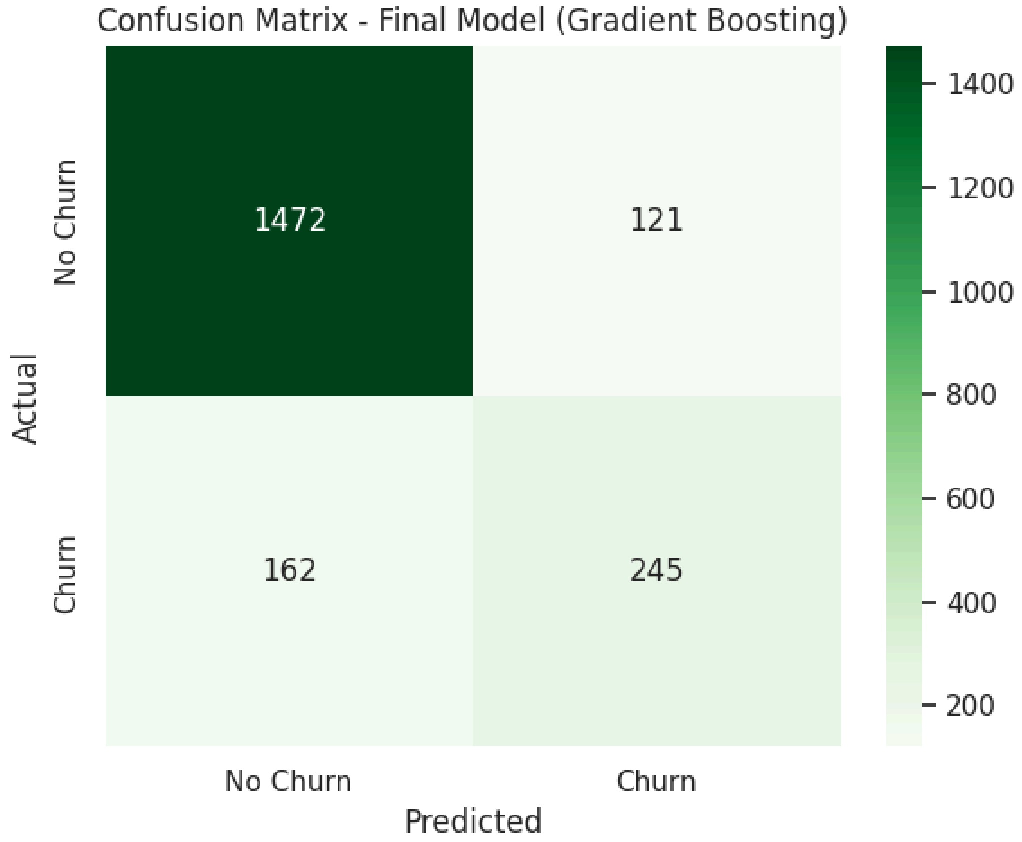
Model	F1	ROC-AUC
Gradient Boosting	0.630	0.863
Random Forest	0.614	0.846
Logistic Regression	0.501	0.774

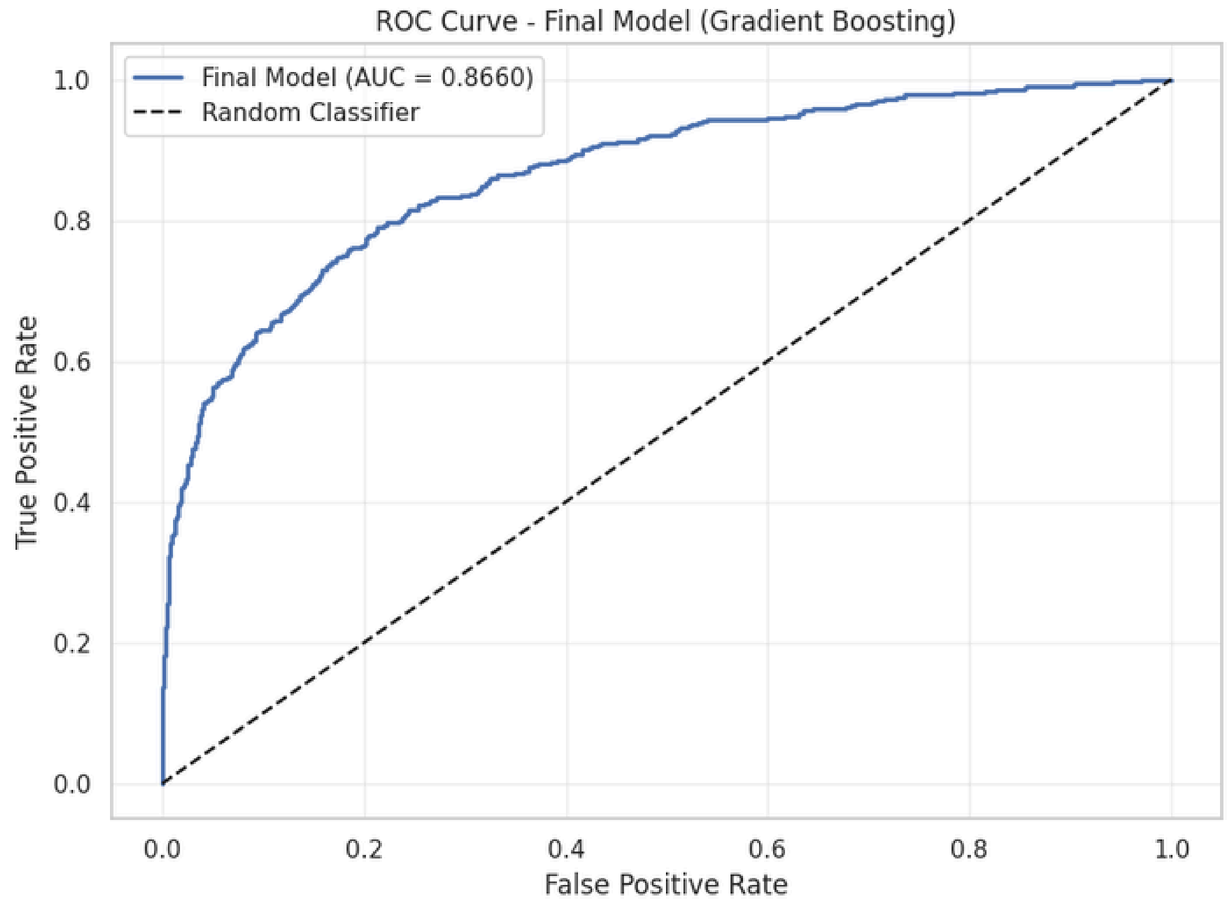
Gradient Boosting wins

Best balance of precision and recall on the minority (churn) class, and the highest ROC-AUC — selected for hyperparameter tuning.

Final Tuned Model

Gradient Boosting after GridSearchCV hyperparameter tuning



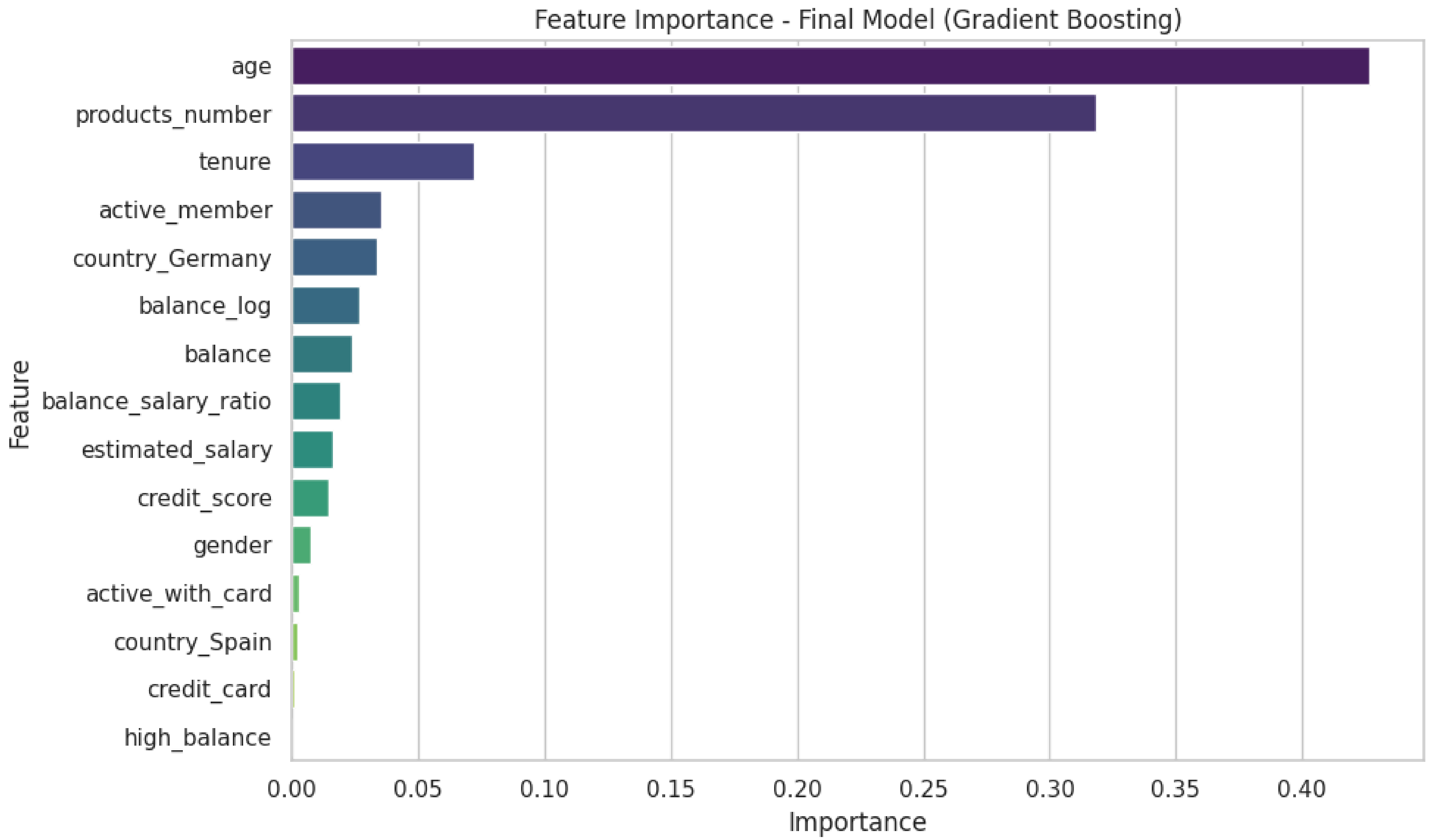


Accuracy	**0.859**
Precision	**0.669**
Recall	**0.602**
F1-Score	**0.634**
ROC-AUC	**0.866**

Best params: learning_rate 0.05, max_depth 5, n_estimators 200

What Drives the Model's Predictions?

Feature importance from the final Gradient Boosting model



TOP 3 FEATURES

Age 42.7%

of total importance

Products number 31.8%

of total importance

Tenure 7.2%

of total importance

Together, age and product holdings explain nearly three-quarters of the model's predictive power.

Deployment:

https://customer-churn-app-u5appmzscpmrzjujq2jwrv5.streamlit.app/

Demo:

https://drive.google.com/file/d/1xz88sSnKmDhUAUdELdQByt8s2npt0swl/view?usp=drive_link

Conclusions & Recommendations

Turning model insight into retention strategy

Target older customers

Older customers churn at a notably higher rate — build proactive retention outreach for the 45+ age segment.

Encourage multi-product adoption

Fewer products correlates with churn — cross-sell and bundle products to deepen the relationship.

Prioritize the German market

Germany shows disproportionately high churn relative to its customer base — investigate local service gaps.

Re-engage inactive members

Inactivity is one of the strongest churn signals — trigger engagement campaigns for dormant accounts.

Final model: Gradient Boosting | ROC-AUC 0.866 | F1-Score 0.634

Thank You